Download Organizer

Planning due November 1, 2026. Final project due November 29, 2026.

1.1 Problem

Class documents, screenshots, and ZIP files can get mixed together in a downloads folder. I want to find what I need without sorting every file by hand.

1.2 Solution

I will write a Bash script that sorts files from a chosen folder into Documents, Images, Archives, and Other. It will show the planned moves first and will not replace existing files.

2.2 Tech stack

Language: Bash on macOS.

Commands: mkdir -p for folders, mv -n for file moves, tr for lowercase extensions, and printf for messages.

Storage: Files in local category folders. The counts reset each time the script runs.

Data source: Files in the folder I choose.

Tools: Terminal, a text editor, Git, GitHub, and Excalidraw.

Checks: Try files with spaces, uppercase or missing extensions, and duplicate names. Check preview and apply modes, empty folders, and a second run. Check that hidden files, subfolders, and symbolic links stay untouched.

2.1 Eisenhower Matrix

Download Organizer - the Do quadrant is the first version.

DO - Important and urgent 1. Sort files from one folder into four categories. 2. Preview the moves. Leave the files and folders unchanged. 3. Move files with --apply. Keep their names and skip existing destination names. 4. Ignore hidden files, subfolders, and symbolic links. Print the planned, moved, skipped, and failed totals.	**SCHEDULE - Important, not urgent** 1. Let the user choose categories and file types. 2. Save a text log of the moves. 3. Add an undo command.
DELEGATE - Not important, urgent 1. Use mkdir -p to create folders. 2. Use mv -n to move files without replacing existing files. 3. Use tr for lowercase extensions.	**DELETE - Not important, not urgent** 1. A graphical app with buttons and windows. 2. Cloud storage and syncing between devices. 3. Automatic file deletion.

Documents: pdf, txt, doc, docx, ppt, pptx. Images: png, jpg, jpeg, gif. Archives: zip, tar, gz. Other: unrecognized types or files with no extension. Treat .PDF and .pdf as the same type.

2.3 Build plan

Each milestone ends with a Git commit.

1. **Nov 1** - Set up the repository, README, matrix, design document, and diagram.

2. **Nov 5** - Read the folder path and options. Add --help and reject invalid input.

3. **Nov 8** - Choose a category from the file extension. Use Other for unknown or missing extensions.

4. **Nov 12** - Add preview mode. Check that it leaves files and folders unchanged.

5. **Nov 16** - Add --apply. Create the category folders, move files, and skip existing destination names.

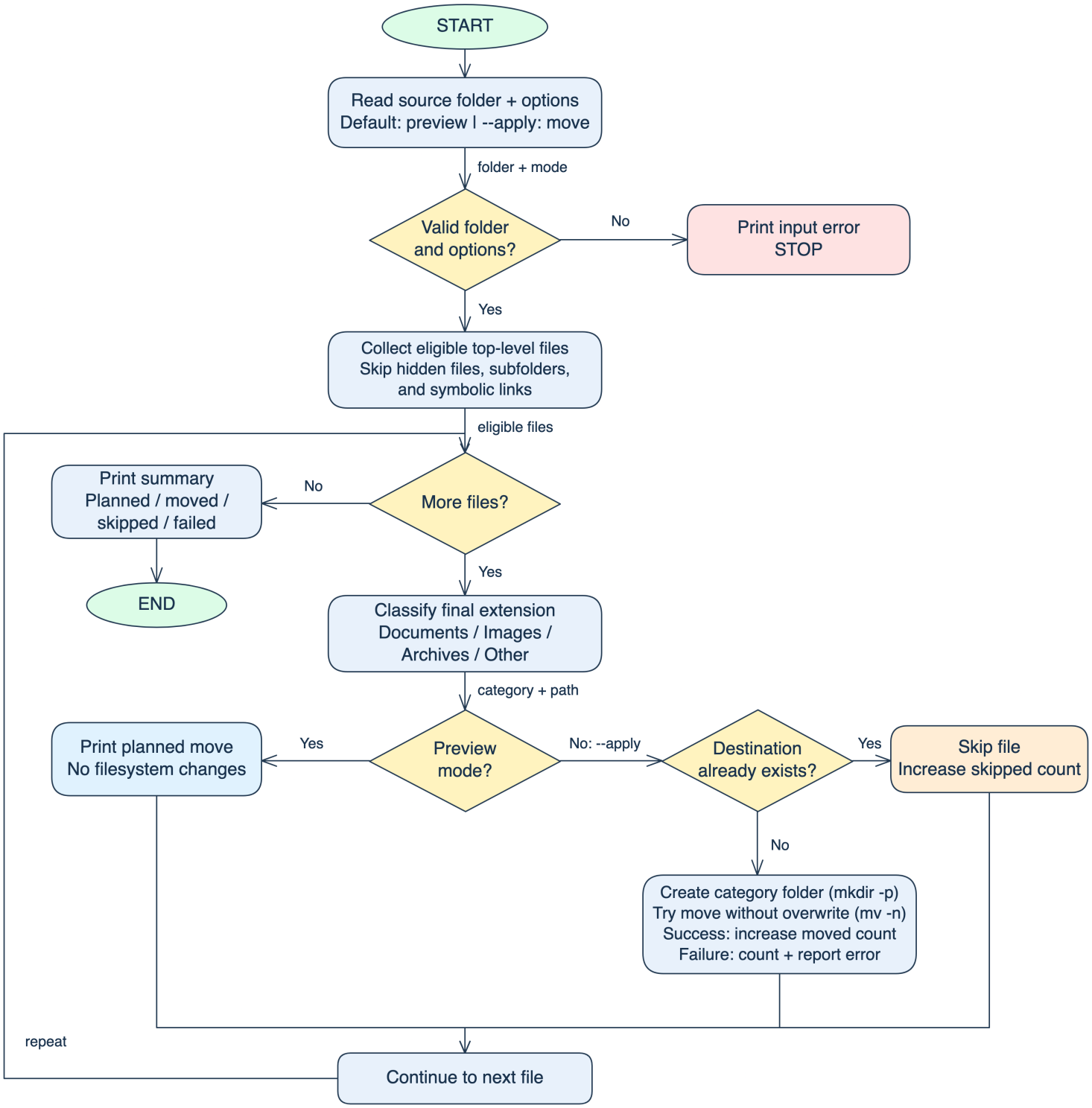
6. **Nov 20** - Print the totals and report skipped files or errors.

7. **Nov 23** - Test spaces, uppercase and missing extensions, duplicate names, empty folders, and repeat runs.

8. **Nov 28** - Update the README and help text. Record the 3-4 minute walkthrough.

Download Organizer

CS 193 - MVP system flow



3.1 Public GitHub repository

https://github.com/neelmd/cs193-download-organizer